COMSATS UNIVERSITY ISLAMABAD,WAH CAMPUS

CIVIL ENGINEERING

SUBJECT: INDETERMINATE STRUCTURAL ANALYSIS

CEP / ASSIGNMENT

SUBMITTED TO: ENGR TASADDIQ RAJA

GROUP MEMBERS:

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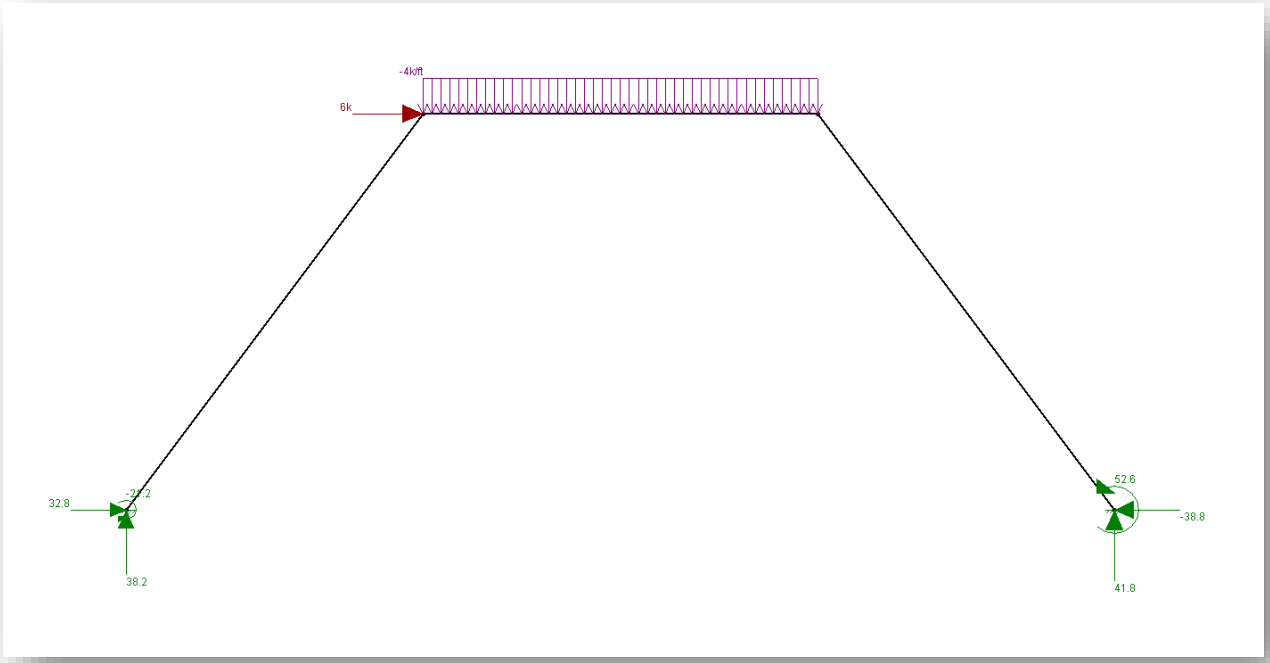
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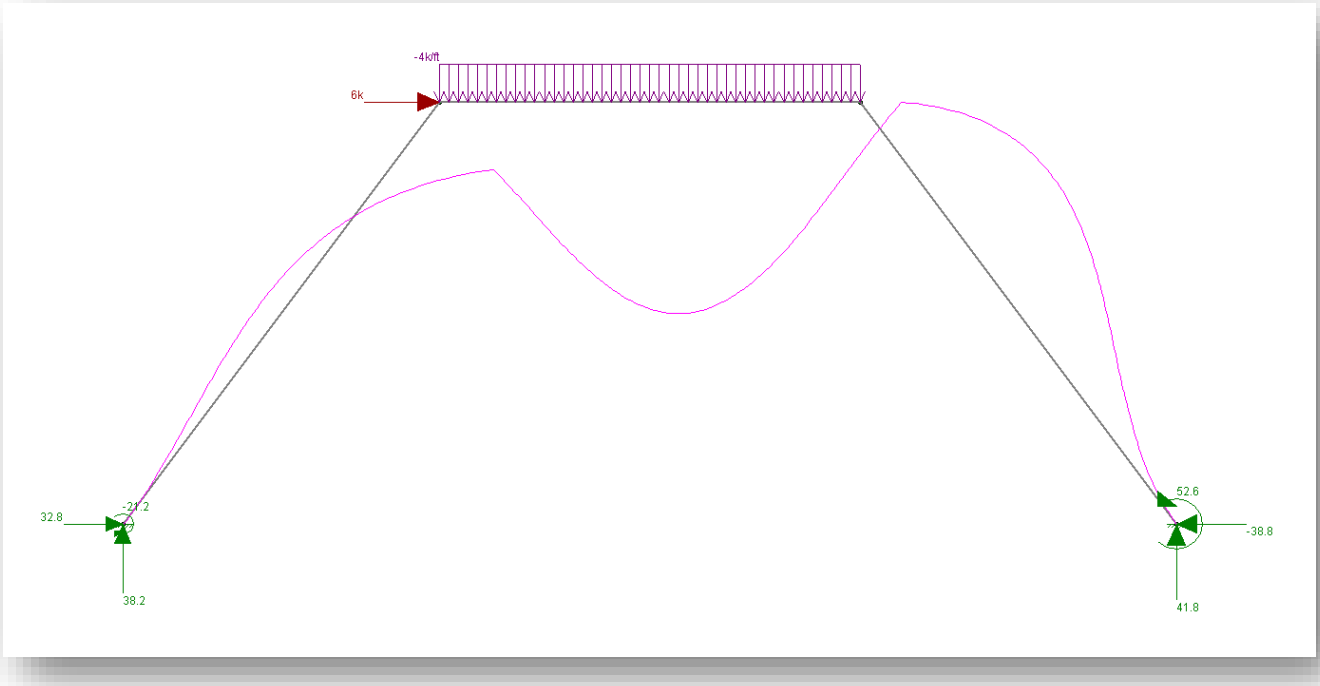
CLASS / SECTION: 5A

DATE:

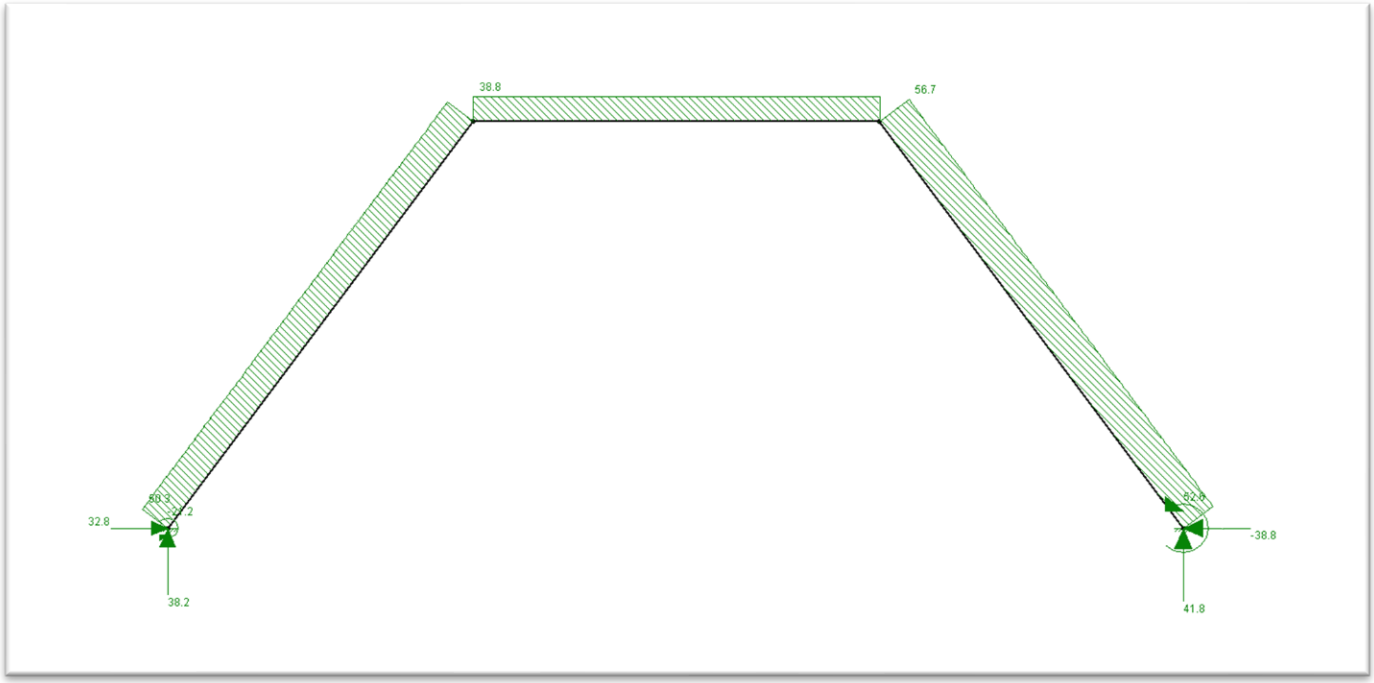
COMPARISON OF SOFTWARE AND MANUAL CALCULATIONS



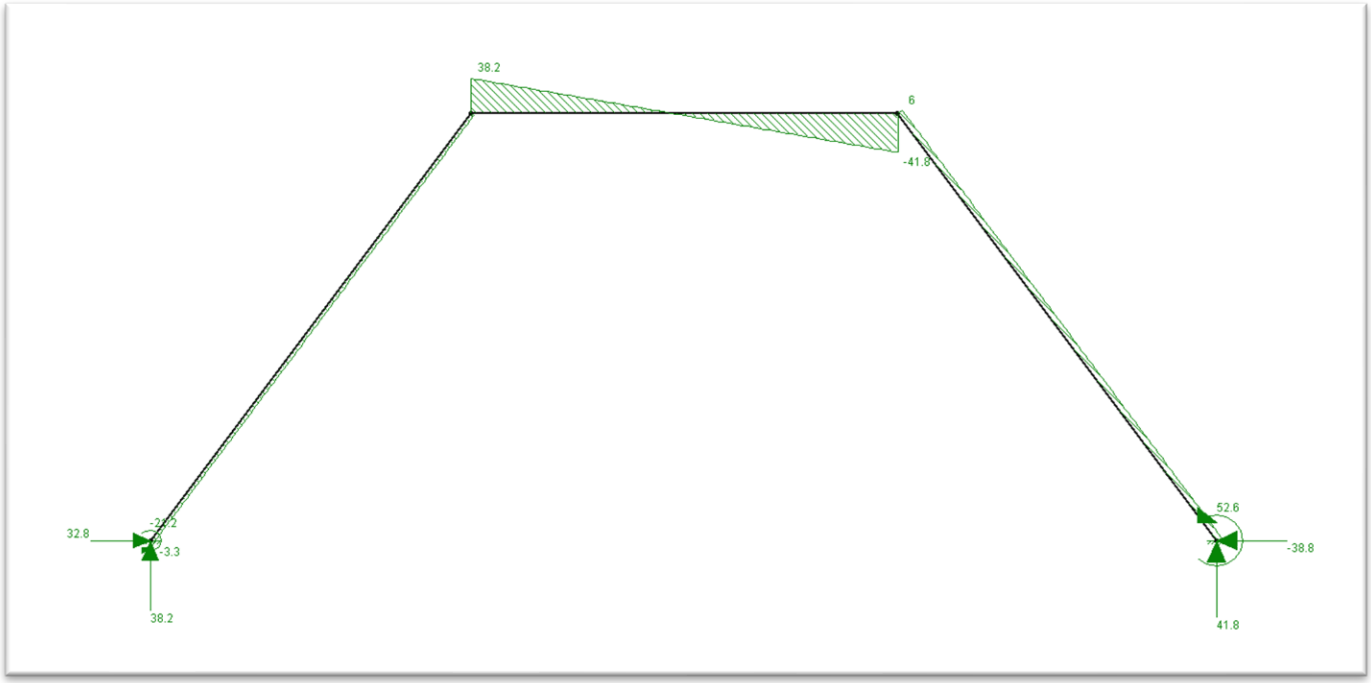
SUPPORT REACTIONS



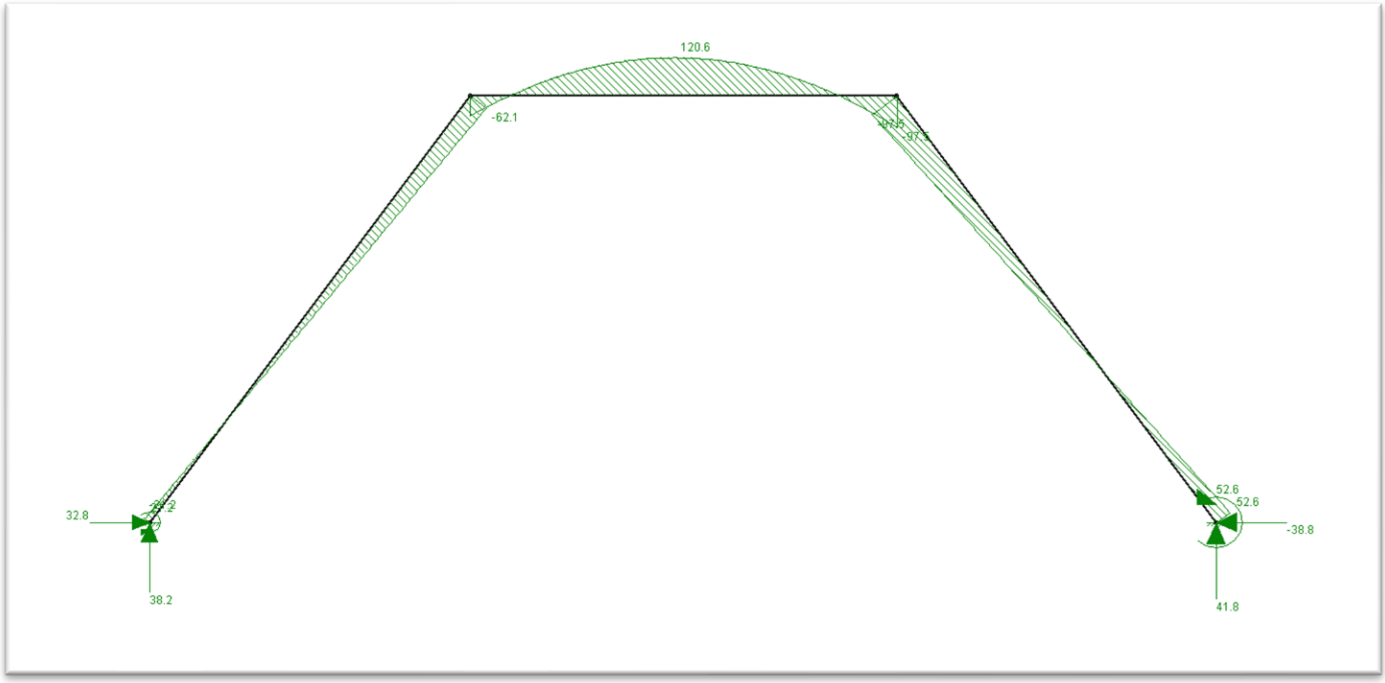
DEFLECTION



AXIAL FORCE DIAGRAM



SHEAR FORCE DIAGRAM



BENDING MOMENT DIAGRAM

The slight variations observed between the manual calculations and the RISA2D results for reactions, axial forces, shear forces, and bending moments are primarily due to differences in the underlying methods used for analysis. RISA2D employs advanced numerical techniques and computational algorithms that account for factors such as load distribution, member stiffness, and boundary conditions with a higher level of precision. In contrast, manual calculations often involve simplified assumptions and approximations, which can lead to minor discrepancies. Additionally, RISA2D's modeling approach may include additional factors like second-order effects or finer discretization, contributing to variations in results. Despite these differences, the overall trends and behavior of the structure are consistent, demonstrating the reliability of both methods for practical purposes.